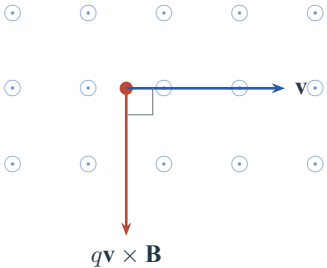
M1-02 磁场力与速度垂直

图示 $q>0$、$\mathbf{B}$ 指向纸外



$q<0$ 时力反向，垂直关系保持不变

$$\mathbf{F}_B = q\mathbf{v}\times\mathbf{B}$$

$$P_B = \mathbf{F}_B\cdot\mathbf{v} = 0$$

$$\frac{d}{dt}\left(\frac{1}{2}mv^2\right) = 0$$

仅就磁场力而言，速率不变。